MathEXplainedMAGAZINE

November Crossword Solutions

Kevin Chen, Wenhao Lu, Aaron Qin

1 **1**	2 **1**
3 **4**	**8**

ONE ACROSS

Coach K can include at most 5 plays, since 6 plays give at least $\binom{6}{1}+\binom{6}{2}+\binom{6}{3}+\binom{6}{4}=56$ combinations of 4 or fewer plays whose totals are at most $14+13+12+11=50$, forcing repeated totals by the Pigeonhole Principle; choosing the largest plays to avoid conflicts, the set $\{7,10,12,13,14\}$ satisfies the rule, yielding a maximum total of $7+10+12+13+14=56$, which cannot be exceeded. Thus, the answer is $5+6=\boxed{11}$.

ONE DOWN

Notice that the data can be modeled with the conditions

$$f(30)=90,$$

$$\frac{1}{10}\sum_{n=21}^{30} f(n)=92.1,$$

$$\frac{1}{20}\sum_{n=11}^{30} f(n)=97.6.$$

Thus, we have the system

$$a(30)^2+b(30)+c=90,$$

$$\sum_{n=21}^{30}(an^2+bn+c)=10\cdot 92.1,$$

$$\sum_{n=11}^{30}(an^2+bn+c)=20\cdot 97.6.$$

Using the identities

$$\sum_{n=m}^{k} n=\frac{k(k+1)}{2}-\frac{(m-1)m}{2},\quad \sum_{n=m}^{k} n^2=\frac{k(k+1)(2k+1)}{6}-\frac{(m-1)m(2m-1)}{6},$$

we evaluate the sums and solve the linear system, obtaining

$$(a,b,c)=(0.05,-3.15,139.5).$$

Hence, the function should be

$$0.05n^2-3.15n+139.5.$$

Therefore, $P=f(1)=0.05-3.15+139.5=136.4$, which gives us an answer of $1+3+6+4=\boxed{14}$.

TWO DOWN

Let $x=\frac{5}{3}, y=\frac{5}{3}, z=2$ be the lengths of the three pairs of opposite edges.

A known formula for the volume V of a disphenoid with opposite-edge lengths x, y, z is

$$72V^2 = (-x^2 + y^2 + z^2)(x^2 - y^2 + z^2)(x^2 + y^2 - z^2).$$

With $x = y = \frac{5}{3}$ and $z = 2$, we can get

$$-x^2 + y^2 + z^2 = 4,$$
$$x^2 - y^2 + z^2 = 4,$$
$$x^2 + y^2 - z^2 = \frac{14}{9}.$$

Thus,

$$72V^2 = 4 \cdot 4 \cdot \frac{14}{9} = \frac{224}{9} \Rightarrow V^2 = \frac{224}{648} = \frac{28}{81} \Rightarrow V = \sqrt{\frac{28}{81}} = \frac{2\sqrt{7}}{9}.$$

Then, $(a, b, c) = (2, 7, 9)$, so the answer is $2 + 7 + 9 = \boxed{18}$.

THREE ACROSS

From the problem statement, the shot is modeled by the parabola

$$y = f(x) = -\frac{1}{3}x^2 + ux + 3, \text{with } u = \tan t,$$

and the hoop is the circle

$$(x-6)^2 + (y-7)^2 = 1.$$

If the ball grazes the hoop at $P = (x_p, y_p)$, the parabola and circle are tangent at P. Thus, P satisfies the circle equation and the equality of slopes (where the derivative of f equals the implicit slope of the circle).

$$A(x, u) := f(x) - 7 = -\frac{1}{3}x^2 + ux - 4, f'(x) = -\frac{2}{3}x + u.$$

The two tangency conditions are

$$(x-6)^2 + A(x, u)^2 = 1, \tag{1}$$

and (from differentiating $(x-6)^2 + (y-7)^2 = 1$ and equating slopes)

$$f'(x) \;=\; -\frac{x-6}{A(x, u)}. \tag{2}$$

Multiply (2) through by $A(x, u)$ to avoid denominators. Writing out $(-\frac{2}{3}x + u) \cdot A(x, u) + (x - 6) = 0$ and expanding yields a quadratic equation in u with coefficients depending on x:

$$xu^2 - (x^2 + 4)u + \left(\tfrac{2}{9}x^3 + \tfrac{11}{3}x - 6\right) = 0. \tag{T}$$

Then, after expanding the circle equation $(x-6)^2 + A(x, u)^2 = 1$, it becomes

$$x^2u^2 - \tfrac{2}{3}x^3u - 8xu + \tfrac{1}{9}x^4 + \tfrac{11}{3}x^2 - 12x + 51 = 0. \tag{C}$$

Equations (T) and (C) are two polynomial equations for the two unknowns u and x. To eliminate u and obtain an equation solely in x, one may compute the resultant of (T) and (C) with respect to u. Carrying out the elimination by forming the resultant or by eliminating u via substitution and clearing denominators produces a polynomial equation of degree 6 in x. Up to an innocuous factor of $x^2/9$, the eliminated polynomial is

$$P(x) := x^6 - 12x^5 + 11x^4 + 288x^3 - 372x^2 - 5508x + 16065 \;= 0.$$

(There is also the double root $x = 0$ coming from the factor x^2; the physical tangency points are among the nonzero roots of P.)

The sextic $P(x)$ has exactly two positive real roots. We must consider the left-hand x_p root because the plotted contact corresponds to the bottom-left.

Newton iteration approach to find the root x_p. Let $F(x) = P(x)$. Newton's method,

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)},$$

converges rapidly for a good initial guess. Start with $x_0 = 5.2$. (A simple plot or sign check of $P(x)$ shows a change of sign near 5.2, so this is a reasonable seed.)

Compute the following iterations:

n	x_n	$F(x_n) = P(x_n)$
0	5.20	48.52742
1	5.2548	0.047112
2	5.2548532	0.000077
3	5.2548532872	5.603537×10^{-12}

Evidently, after 3 Newton iterations, we already have

$$x_p \approx 5.2548532872.$$

The problem asks for the last two digits of $\lfloor 10000 \cdot x_p \rfloor$, so compute

$$10000 \cdot x_p \approx 52548.532872 \Rightarrow \lfloor 10000 \cdot x_p \rfloor \bmod 100 \Rightarrow \boxed{48}.$$